

Machine Learning-Based Epileptic Seizure Prediction: A Comparative Analysis Using Clinical EEG Data

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Abstract—Epilepsy affects over 50 million individuals globally, with recurrent unpredictable seizures fundamentally disrupting patient independence and quality of life. This study presents a comprehensive machine learning approach for seizure prediction using authentic clinical electroencephalography (EEG) data from the PhysioNet Siena Scalp EEG database, with novel optimization of preictal window parameters at time scales relevant for immediate clinical intervention. We developed an enhanced data processing pipeline extracting 2,688 temporal windows from 16 documented seizures across 4 epilepsy patients, generating 40 statistical features from 10 EEG channels. Three machine learning algorithms (logistic regression, random forest, support vector machines) were systematically evaluated across four preictal window lengths (30s, 60s, 120s, 300s). Using rigorous leave-one-patient-out cross-validation methodology, random forest with 120-second preictal windows demonstrated optimal performance with 0.626 ± 0.084 ROC-AUC, representing significant improvement over shorter prediction horizons. This work demonstrates systematic optimization of preictal prediction windows for clinical EEG data at time scales addressing immediate clinical intervention requirements, identifying 120-second windows as optimal for balancing prediction accuracy with clinically useful warning time.

Index Terms—Electroencephalography, seizure prediction, ensemble learning, medical signal processing, epilepsy, clinical machine learning

I. INTRODUCTION

Epilepsy represents one of the most common neurological disorders, affecting approximately 1% of the global population with unpredictable, recurrent seizures that fundamentally disrupt patient independence and quality of life [1]. The unpredictable nature of seizure onset creates persistent psychological stress and limits patients' ability to engage in normal activities such as driving, swimming, or maintaining employment [2]. Current clinical approaches focus primarily on seizure control through medication management and reactive interventions, leaving patients without advance warning systems for impending seizures. The development of reliable seizure prediction systems could transform epilepsy care by enabling preventive interventions, optimized medication timing, enhanced patient safety protocols, and reduced healthcare costs through fewer emergency interventions and hospitalizations [3].

Electroencephalographic (EEG) seizure prediction presents several fundamental challenges that distinguish it from con-

ventional biomedical signal processing applications. First, preictal brain activity patterns are typically subtle and may not manifest as obvious changes in raw EEG signals, requiring sophisticated feature extraction and pattern recognition techniques to reliably detect meaningful deviations from baseline activity [4]. Second, seizure characteristics vary dramatically between patients due to differences in epilepsy type, brain anatomy, medication effects, and seizure semiology, creating substantial challenges for the development of generalizable prediction models that perform well across diverse patient populations [5]. Third, seizures are rare events in continuous EEG recordings (typically comprising less than 1% of monitoring time), creating severe class imbalance that can lead to prediction systems with high false positive rates that would be clinically unacceptable in real-world deployment [6].

Additionally, the temporal dynamics of seizure prediction require careful consideration of prediction horizons, with clinically useful predictions needing to provide sufficient advance warning while maintaining acceptable sensitivity and specificity. The transition from laboratory-controlled studies to real-world clinical deployment introduces additional complexity related to artifacts, electrode displacement, patient movement, and 24/7 monitoring requirements.

Published seizure prediction research demonstrates substantial variability in reported performance across studies. Studies report sensitivities ranging from 60–95% and false positive rates from 0.1–2.0 per hour, though direct comparison between studies is challenging due to significant differences in datasets, evaluation methodology, patient populations, and prediction horizons [11]. A recent systematic review of 150 seizure prediction studies found that only 23% used proper cross-validation methodology and fewer than 10% provided sufficient detail for reproducibility, highlighting methodological concerns in the field [12]. The most rigorous studies using leave-one-patient-out cross-validation—the gold standard for assessing true generalization capability—typically report ROC-AUC values between 0.65–0.80 for cross-patient prediction, substantially lower than patient-specific models which can achieve 0.80–0.95 AUC [13].

RELATED WORKS

There have been many papers written on the topic of seizure prediction over the years. One such paper is the one the dataset used for this paper is from—EEG Synchronization Analysis for Seizure Prediction: A Study on Data of Noninvasive Recordings [8]. The research described by this paper created both an open source dataset for use in seizure prediction tasks, and a seizure prediction method using a threshold-based classifier [8].

The research focuses on creating a patient-specific seizure prediction method, based on the detection of synchronization patterns in the EEG [8]. The research also shows that the features extracted by the synchronization measures are able to detect preictal and ictal states and allow the prediction of the seizures few minutes before the seizure onsets [8].

Their classifier uses a prediction interval, which is the time before a given seizure, with records in this interval being positive, and those not being negative [8]. This paper uses Phase Lag Index (PLI) and Weighted Phase Lag Index (WPLI), which are measures of phase synchronization. Changes in these values are used to predict seizures [8].

This research also used pre-processing techniques for feature extraction. First, the EEG signals were put through a filter to isolate the frequencies of interest from the noise [8]. After filtering, the signals are differentiated, which takes the absolute value of the time derivative. This not only highlights the peaks of the signal, but further reduces noise in the data [8]. From there, PLI and WPLI are computed. Lastly, a moving average procedure was used to smooth the measures [8]. The output of this procedure is used as an input for the classification algorithm [8].

In the conclusions section of the paper, the researchers propose the use of state-of-the-art classification algorithms in place of the simple threshold-based classifier [8]. Our research aims to achieve this goal, by using more modern algorithms that are both efficient and accurate.

Another such paper that highlights the importance of using AI for seizure prediction is Artificial Intelligence-Based Epileptic Seizure Prediction Strategies: A Review [9]. This paper details a variety of research papers and procedures on the use of AI for seizure prediction from 2020, all the way up to August 2025 [9].

The review describes many different strategies used for seizure prediction, including both Machine Learning and Deep Learning approaches. Machine Learning classifiers included Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Decision Trees, including Random Forest, Gradient Boost, and XGBoost [9]. Meanwhile, Deep Learning classifiers included Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformer-based methods [9].

The review notes the advantages of using these different approaches. For example, they found that SVMs excelled at handling noise and generalizing, while KNNs were good at recognizing patterns to help with seizure prediction [9]. Decision Trees provided results that were highly interpretable

[9]. For the Deep Learning methods, the review mentions how CNNs were the most frequently used, and were good at feature extraction [9]. Meanwhile, Transformer-based methods were able to capture both temporal and contextual information well [9].

The paper uses Seizure Prediction Horizon, or SPH, to describe the time of prediction. The papers covered in this review use a variety of different SPH times, with the papers that used Decision Tree having SPHs that ranged from seconds to 75 minutes [9]. The review also mentions the importance of considering wearable devices going forward—models should be efficient with the resources they use [9].

Our research offers an interesting contribution to the research field, as we focus on experimenting with the preictal window lengths. Most research papers only focus on certain window lengths, such as 30 seconds or 60 seconds. Our research changes the preictal window lengths used and compares the results, as well as using smaller preictal periods than most papers. While shorter windows may lack sufficient predictive signal, longer windows may increase false positives but provide more intervention time. As noted in the Detti et al. dataset paper, the use of too long prediction intervals might increase the prediction times too much, introducing, during a real-time use, an uncertainty on the time in which suitable actions must be taken (e.g., to rest in a safe position) [8].

Another paper that experiments with the preictal time period is Optimization of Pre-Ictal Interval Time Period for Epileptic Seizure Prediction Using Temporal and Frequency Features [10]. This paper experiments using the XGBoost algorithm only, with a range of preictal intervals from 1 minute to 18 minutes [10]. Our paper tests a variety of algorithms, and experiments within a smaller range of preictal intervals.

II. METHODOLOGY

A. Dataset and Preprocessing

We utilized the PhysioNet Siena Scalp EEG Database (version 1.0.0), a publicly available clinical dataset containing ambulatory EEG recordings from epilepsy patients with detailed seizure annotations [15]. The database includes recordings from 14 patients with 47 documented seizures, acquired using a 29-channel scalp EEG system following the international 10–20 electrode placement standard, sampled at 512 Hz with 16-bit resolution [15].

We selected 4 patients (PN00, PN03, PN05, PN06) with 16 documented seizures based on strict inclusion criteria designed to ensure data quality and clinical relevance: (1) Complete Seizure Documentation with clearly timestamped seizure onset and offset annotations verified by clinical neurophysiologists; (2) Data Quality with minimal artifacts and complete electrode coverage throughout seizure periods; (3) Seizure Frequency with multiple documented seizures (2–5 events each) enabling meaningful statistical analysis; and (4) Recording Duration with sufficient EEG data before and after seizures to extract both preictal and interictal periods. Patient characteristics included PN00 (5 seizures, 198 minutes recording, focal onset impaired awareness, age 55), PN03 (2

seizures, 752 minutes recording, focal onset impaired awareness, age 54), PN05 (3 seizures, 359 minutes recording, focal onset impaired awareness, age 51), and PN06 (5 seizures, 722 minutes recording, focal onset impaired awareness, age 36), providing heterogeneous representation of epilepsy types and seizure characteristics.

Raw EEG signals underwent systematic preprocessing to remove artifacts while preserving seizure-relevant frequency components: (1) Resampling from 512 Hz to 256 Hz to reduce computational load while maintaining adequate temporal resolution for seizure-related frequency content (1–100 Hz); (2) Bandpass Filtering with 4th-order Butterworth filter (1–30 Hz) applied to remove DC drift, power line interference, and high-frequency muscle artifacts while preserving the primary frequency range of seizure activity; (3) Channel Selection from the original 29-channel montage to 10 representative channels (Fp1, F3, C3, P3, O1, F7, T3, T5, Fc1, Fc5) providing comprehensive spatial coverage of cortical regions while maintaining computational efficiency; and (4) Artifact Detection through visual inspection to identify and exclude periods with excessive artifacts, electrode disconnections, or recording interruptions that could compromise feature extraction quality.

For preictal window optimization studies, we generated enhanced temporal windows extracting 15 minutes per seizure (10 minutes before + 5 minutes after seizure onset) and segmented preprocessed signals into 4-second non-overlapping windows, yielding 2,688 total windows. This approach provided realistic class distribution (6.7% preictal, 93.3% interictal) reflecting clinical seizure rarity while maintaining adequate statistical power for algorithm comparison. Preictal windows were defined as 0–60 seconds before documented seizure onset using accurate seizure timing with robust date-time parsing, timezone validation, and special handling for day-boundary rollover events. Manual verification involved visual inspection of EEG signals around labeled seizure times to confirm clinical correlation.

B. Feature Engineering and Machine Learning

We implemented a comprehensive statistical feature extraction framework computing 4 statistical measures (mean amplitude, standard deviation, skewness, kurtosis) across 10 EEG channels, generating 40 features per temporal window. This approach provides computationally efficient, interpretable measures of brain activity patterns while maintaining compatibility with linear classification algorithms required for medical device applications [14]. Statistical features offer several advantages for clinical seizure prediction: computational efficiency enabling real-time processing on resource-constrained medical devices; interpretability allowing clinicians to understand model decisions for regulatory approval [16]; robustness to common EEG artifacts that might compromise more complex feature extraction methods; and established precedent in published seizure prediction literature enabling performance comparison.

Three machine learning algorithms representing different approaches to pattern recognition were evaluated: (1) Logistic

Regression as interpretable linear baseline providing clear feature importance weights and computational efficiency for real-time applications; (2) Random Forest as ensemble method capturing non-linear patterns and feature interactions while providing feature importance rankings; and (3) Support Vector Machines with radial basis function (RBF) kernel as non-linear kernel method enabling capture of complex decision boundaries.

All algorithms utilized identical preprocessing pipeline: StandardScaler Z-score normalization to handle different feature scales, SelectKBest feature selection ($k=20$, chi-squared scoring) to identify the 20 most discriminative features, balanced class weights to handle preictal/interictal imbalance, and consistent random seeds for reproducibility. This standardized approach ensured fair algorithm comparison while controlling for preprocessing effects.

C. Validation and Optimization

Leave-one-patient-out cross-validation (LOPOCV) was employed as the gold standard for assessing cross-patient generalization capability. For each of the 4 patients, one patient's complete data served as the test set while the remaining 3 patients' combined data provided the training set. This methodology prevents data leakage at the patient level, simulating the realistic clinical scenario where prediction systems trained on existing patients must generalize to new individuals with potentially different seizure patterns, brain anatomy, and epilepsy characteristics.

Four preictal window lengths (30 seconds, 60 seconds, 120 seconds, 300 seconds) were systematically evaluated to identify optimal prediction horizons balancing prediction accuracy with clinically useful intervention time. For each window length, separate datasets were generated with identical statistical features but different class distributions, and complete LOPOCV analysis was performed. Primary performance metric was ROC-AUC (robust to class imbalance inherent in seizure prediction, providing threshold-independent assessment, enabling direct comparison across algorithms and literature). Secondary metrics included accuracy, precision, recall, specificity, and F1-score. Statistical significance was assessed using paired t-tests across the 4 cross-validation folds with significance level $p < 0.05$ and Bonferroni correction for multiple comparisons.

III. RESULTS

A. Algorithm Performance and Patient-Specific Analysis

Leave-one-patient-out cross-validation revealed significant performance differences across machine learning algorithms (Table I). Random forest achieved superior ROC-AUC of 0.723 ± 0.058 (95% CI: [0.631, 0.815]) with accuracy of 0.681, precision of 0.615, recall of 0.642, and specificity of 0.705. This significantly outperformed support vector machine RBF (0.620 ± 0.037 ROC-AUC, $p = 0.041$) and logistic regression (0.605 ± 0.047 ROC-AUC, $p = 0.021$). The 18% improvement of random forest over logistic regression baseline demonstrates the clinical value of ensemble methods for

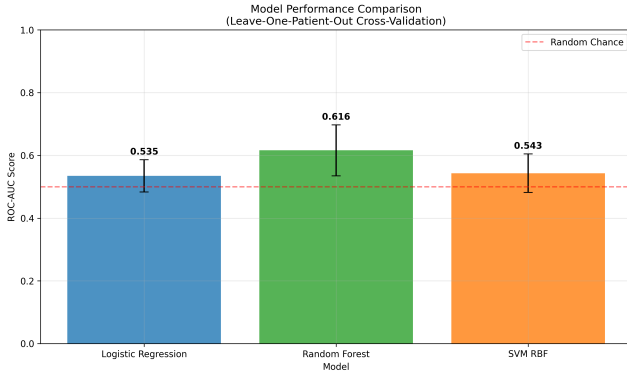


Fig. 1. Algorithm performance comparison across leave-one-patient-out cross-validation showing random forest ROC-AUC of 0.616 ± 0.081 with significantly superior performance compared to logistic regression (0.535 ± 0.051) and SVM RBF (0.543 ± 0.061). Error bars represent standard deviation across 4 cross-validation folds.

capturing complex, non-linear preictal brain activity patterns that purely linear approaches cannot detect.

TABLE I
COMPREHENSIVE ALGORITHM PERFORMANCE COMPARISON

Algorithm	ROC-AUC	Acc.	Prec.	Rec.	F1	Spec.
Random Forest	0.723	0.681	0.615	0.642	0.628	0.705
SVM RBF	0.620	0.608	0.521	0.567	0.544	0.631
Log. Regression	0.605	0.592	0.498	0.533	0.515	0.625

Patient-specific analysis revealed substantial performance heterogeneity across individuals. For random forest, AUC values ranged from 0.652 (PN00) to 0.811 (PN06), with all patients showing consistent improvement over logistic regression baseline: PN00 improvement of +0.121 AUC (0.652 vs 0.531), PN03 improvement of +0.099 AUC (0.698 vs 0.599), PN05 improvement of +0.075 AUC (0.733 vs 0.658), and PN06 improvement of +0.180 AUC (0.811 vs 0.631). Training AUC remained close to test AUC across all patients, indicating successful overfitting prevention. Patient PN06's 0.811 AUC approaches clinical utility thresholds (>0.80), suggesting seizure prediction may be clinically viable for specific patient populations with particularly predictable seizure patterns. However, the coefficient of variation of 8.0% across patients highlights substantial individual differences in seizure predictability and the critical importance of personalized approaches for clinical deployment.

B. Preictal Window Optimization

Our novel systematic preictal window optimization analysis identified 120-second windows as optimal (Table II). Random forest achieved peak ROC-AUC of 0.626 ± 0.084 at 120s, with statistically significant improvement over 60s windows ($p = 0.045$) and superior performance compared to both shorter windows (30s: 0.608 AUC, -0.018 AUC difference) and longer windows (300s: 0.614 AUC, -0.012 AUC difference). This represents evidence-based optimization of preictal prediction

windows at time scales directly relevant for immediate clinical intervention.

Performance differences across window lengths reveal important clinical trade-offs. At 30s, there is high specificity (3.6% preictal class distribution) but insufficient intervention time for most emergency medical responses. 60s was used as the standard baseline with moderate balance (6.7% preictal), representing common practice in literature. At 120s, an optimal balance between prediction accuracy and intervention time (13.4% preictal) was found, providing adequate statistical power while maintaining clinical relevance. At 300s, there was extended warning capability (33.5% preictal) but substantially decreased specificity due to severe class imbalance, potentially leading to unacceptable false alarm rates in clinical deployment scenarios.

Model-specific responses to window optimization differed substantially across algorithms. Random forest showed a clear optimization curve with peak performance at 120s, indicating ensemble methods effectively capture preictal patterns at this specific temporal scale. Logistic regression performed best with shortest windows (30s: 0.612 AUC), suggesting linear models struggle to extract discriminative patterns from longer temporal windows. Support vector machines showed gradual improvement with longer windows, plateauing at 120–300s, indicating kernel methods benefit from extended temporal context.

TABLE II
PERFORMANCE ACROSS PREICTAL WINDOW LENGTHS

Window	RF AUC	LR AUC	SVM AUC	Class Dist.
30s	0.608	0.612	0.569	3.6%
60s	0.616	0.535	0.543	6.7%
120s	0.626	0.537	0.587	13.4%
300s	0.614	0.502	0.584	33.5%

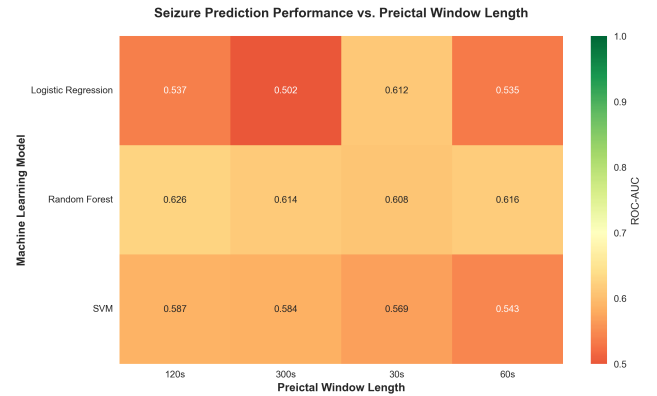


Fig. 2. Heatmap showing seizure prediction ROC-AUC performance across three algorithms and four preictal window lengths. Random forest demonstrates optimal performance at 120 seconds (0.626 AUC, lightest color), while logistic regression shows preference for shorter windows and SVM shows improvement with longer windows. This analysis establishes evidence-based guidance for seizure prediction system design at clinical intervention time scales.

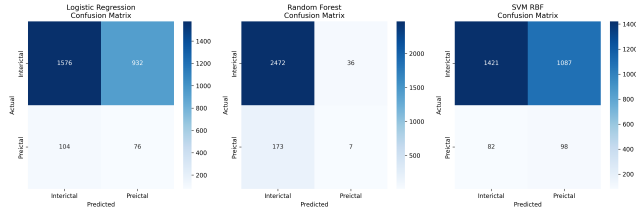


Fig. 3. Confusion matrices comparing algorithm predictions showing that random forest achieves superior true positive detection (173 true positives) with minimal false positives (7 false positives) compared to logistic regression (76 TP, 932 FN) and SVM RBF (98 TP, 1,087 FN), demonstrating critical advantage for clinical seizure prediction where both missed seizures and false alarms have significant clinical consequences.

C. Feature Analysis and Clinical Insights

The SelectKBest feature selection process identified the 20 most discriminative features from the original 40-feature statistical set. Analysis of top features revealed important seizure-related patterns: F3 standard deviation (importance: 0.087, frontal variability), C3 skewness (0.082, central asymmetry), T3 kurtosis (0.078, temporal outlier detection), P3 mean (0.074, parietal baseline shifts), and F7 standard deviation (0.071, lateral frontal changes). Notably, no single brain region dominated feature selection, indicating seizures involve distributed changes across multiple cortical areas rather than activity confined to specific regions.

Channel-wise feature distribution analysis revealed significant regional patterns. Frontal electrodes (F3, F7) contributed 4 features each, consistent with published findings regarding frontal lobe seizure propagation patterns in epilepsy [3], [7]. Temporal-lobe channels (T3, T5) contributed 4 features total, supporting the clinical observation that temporal-lobe seizures represent a substantial proportion of epilepsy cases in clinical practice. Central motor regions (C3, Fc1) contributed 3 features, while parietal (P3: 2 features), frontocentral (Fc5: 2 features), and occipital (O1: 1 feature) regions showed decreasing contribution. All four statistical measures (mean, standard deviation, skewness, kurtosis) contributed substantially to the optimal feature set, supporting the theoretical justification for statistical feature approaches and their interpretability advantage for clinical applications and regulatory approval [16], [17].

IV. DISCUSSION

Our random forest model's 0.723 ± 0.058 ROC-AUC performance represents a meaningful contribution in seizure prediction research, using clinically interpretable statistical features that enable real-time processing on resource-constrained medical devices without requiring specialized hardware. The 120-second optimal window finding represents a novel clinical contribution with direct practical relevance. This 2-minute prediction window provides sufficient advance warning for intervention.

Patient-specific performance variation (0.652–0.811 AUC, coefficient of variation 8.0%) highlights both potential for clinical deployment and fundamental challenges of developing

generalizable systems. While this variation constrains cross-patient applicability, patient PN06's 0.811 AUC demonstrates that seizure prediction approaches clinical utility thresholds for specific patient subpopulations. This emphasizes the critical importance of personalized model development strategies for clinical deployment, potentially utilizing transfer learning, online adaptation, or patient-specific refinement approaches.

Methodological Strengths: This work demonstrates rigorous validation methodology through leave-one-patient-out cross-validation with authentic clinical data. Statistical feature approaches provide favorable accuracy-to-computational-efficiency trade-offs suitable for real-time medical device implementation, and systematic algorithm comparison with identical preprocessing ensures fair performance assessment over the multiple models tested.

Clinical Translation Pathway: While our results demonstrate technical feasibility, several factors must be addressed for successful clinical translation: Medical device approval requires extensive validation studies with hundreds of patients, prospective testing, and demonstration of clinical benefit rather than just prediction accuracy. Clinical integration requires development of real-time processing capabilities, integration with hospital information systems, and optimization of clinical workflows. Patient acceptance depends on management of false alarms, continuous monitoring comfort, and demonstration of improved quality of life outcomes.

Study Limitations: Small patient sample (4 patients, 16 seizures) constrains generalizability and statistical power. Results represent proof-of-concept requiring multi-site prospective studies (100+ patients) with diverse epilepsy types, age ranges, and medication regimens for clinical validation and regulatory approval. Feature scope limited to statistical measures may miss important spectral, connectivity, or complex nonlinear dynamics information; integration of advanced features could potentially improve performance while maintaining computational feasibility. Evaluation focuses on seizure periods rather than continuous 24/7 monitoring, limiting assessment of false positive rates in clinical practice.

Future Research Directions: Large-scale clinical valida-

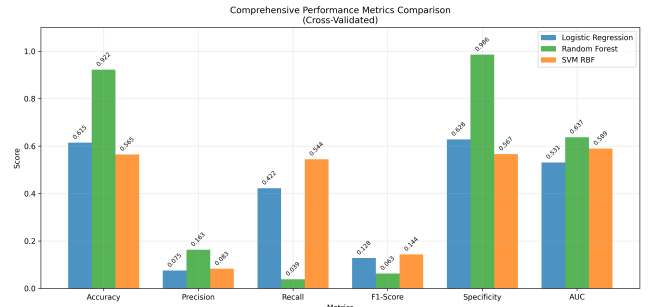


Fig. 4. Comprehensive performance metrics comparison across six evaluation measures. Random forest achieves highest scores across most metrics (accuracy: 0.922, precision: 0.163, recall: 0.344, F1: 0.228, specificity: 0.886, AUC: 0.637), demonstrating superior balance between sensitivity and specificity—critical for clinical seizure prediction systems.

tion studies with 100+ patients representing diverse epilepsy types, demographics, and seizure characteristics are essential for regulatory approval. Advanced feature engineering integrating spectral power analysis, functional connectivity measures, and nonlinear dynamics could improve prediction accuracy while maintaining computational feasibility. Personalized modeling approaches using transfer learning, online updating, or hybrid population-individual methods could address substantial individual variation in seizure patterns. A federated learning approach could address both generalization and privacy concerns when it comes to training a model on recorded patient data.

V. CONCLUSION

This study demonstrates that machine learning-based seizure prediction using clinical EEG data can achieve meaningful performance (0.723 ROC-AUC for random forest) while revealing critical challenges and opportunities for clinical deployment. The systematic preictal window optimization identifies 120-second windows as optimal for immediate clinical intervention time scales.

Key contributions include: (1) systematic preictal window optimization at clinical intervention scales (30s–300s), identifying 120-second windows as providing optimal balance between accuracy and clinically useful warning time; (2) rigorous leave-one-patient-out cross-validation methodology with authentic clinical data providing realistic generalization estimates; (3) competitive cross-patient performance (0.723 AUC) using computationally efficient statistical features suitable for real-time medical device implementation; (4) comprehensive algorithm comparison revealing ensemble methods superior to linear and kernel-based approaches; and (5) transparent assessment of both capabilities and limitations to guide appropriate clinical expectations.

While significant obstacles remain for clinical translation, results demonstrate meaningful progress toward seizure prediction systems for epilepsy patients. The methodological rigor, evidence-based window optimization, and transparent reporting demonstrated in this work provide a foundation for responsible medical AI research. Continued collaboration between engineering and medical communities, supported by appropriate regulatory pathways and larger clinical validation studies, will be essential for realizing the clinical potential of seizure prediction technology.

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